Spacetime Vortex Gravity (SVG) A conceptual model proposing that gravity arises from rotational vortex structures in spacetime

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Abstract

This paper proposes the Spacetime Vortex Gravity (SVG) model, a conceptual framework suggesting that gravity arises not merely from the presence of mass-energy, but from the rotational influence of that mass on the surrounding spacetime. Unlike general relativity, which describes mass as bending spacetime but does not explain why this bending occurs, the SVG model introduces a physical mechanism: spinning mass generates vortical distortions in spacetime, and these rotational effects are the true origin of what we perceive as gravitational attraction.

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1. Introduction

Einstein's general theory of relativity revolutionized our understanding of gravity by describing it as the curvature of spacetime caused by mass and energy. However, general relativity mathematically describes what happens but not the underlying physical cause for why mass curves space. It simply postulates the curvature as a consequence of the energy-momentum tensor in Einstein's field equations.

The Spacetime Vortex Gravity (SVG) model seeks to go deeper. It suggests that the curvature of spacetime is not a passive effect of mass, but an active result of its rotational influence — that is, the spinning of massive bodies creates vortex-like structures in the spacetime fabric. These vortices are responsible for the bending of space and the motion of other objects within it.

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2. Core Concept of SVG

At the heart of SVG lies the idea that massive bodies twist spacetime around them when they spin, much like a spoon spinning in honey creates swirls. This spinning induces rotational strain or "drag" in the surrounding spacetime, forming vortices that act as gravitational wells. Other objects, caught in these vortex fields, follow curved paths — what we interpret as orbits or gravitational attraction.

This concept builds naturally on the known phenomenon of frame dragging (Lense-Thirring effect), where rotating masses slightly twist nearby spacetime. SVG proposes that this effect is not just a minor correction — it is the primary cause of gravity itself.

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3. Analogies and Supportive Evidence

A useful analogy is a spinning ball dropped into a thick fluid. If the ball is still, it sinks with minimal disturbance. But if it spins, it creates whirlpools around it. Similarly, spinning mass doesn’t just sit in spacetime — it stirs it.

In physics, rotating black holes (described by the Kerr metric) already show that spin affects gravity. The stronger the spin, the more spacetime is twisted. SVG generalizes this idea: all gravity originates from such twist, not just in extreme cases.

Moreover, this model is visually intuitive and aligns with observations of spinning galaxies, accretion disks, and the rotational dynamics of cosmic structures — all suggesting that rotation plays a key role in shaping the universe.

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4. Implications of SVG

Causal Explanation: SVG provides a physical reason why space bends — mass induces rotational strain.

Unified Thinking: It aligns with quantum physics, where spin is fundamental, offering a possible path toward quantum gravity.

Testable Ideas: Predictions may include enhanced gravitational effects around spinning objects or new ways to model orbit behavior using vortex math.

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5. Future Work

This concept paper is the first step in building a full theory of Spacetime Vortex Gravity. Future research should:

Extend Einstein's equations to include rotational tensor terms.

Model spacetime as a dynamic medium capable of vortex behavior.

Compare SVG predictions with gravitational wave data and frame-dragging experiments.

SVG may also be expanded to explore connections to inertia, time dilation, and even consciousness if the vortical structure of spacetime has broader implications.

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Conclusion

The SVG model offers a bold and intuitive explanation for gravity, not as an abstract bending of space, but as a physical, vortex-driven phenomenon. It seeks to complete Einstein’s vision by answering the fundamental question: Why does space bend? The answer, according to this theory, lies in the spin.

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